



Medical Microbiology

Third stage- College of Dentistry

Lab. 9

Corynebacterium

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Learning Objectives:

After completing this lab, you should be able to:

1-Describe the microscopic appearance of *Corynebacterium*.

2-Identify the major virulence factors of *Corynebacterium*.

3-Recognize its cultural characteristics and the selective media used for its isolation.

4-Explain the laboratory methods used to detect diphtheria toxin.



Morphology of *Corynebacterium*

- Gram-positive (stain purple/blue).
- Bacilli (rods) they are characteristically club-shaped (tapering at one end).
- Aerobic or facultatively anaerobic
- Non-motile.
- Non-spore forming.
- Non-capsulated.



Arrangement:

Corynebacterium typically shows a “**Chinese-letter**” arrangement, appearing in **palisades** (side-by-side) or forming **V- and Y-shaped** configurations.

Volutin (Metachromatic) Granules:

These bacteria contain intracellular cytoplasmic inclusions known as **Volutin Granules** or **Babes–Ernst Granules**, which serve as **polyphosphate storage** deposits. When stained with **Albert’s stain** or **Neisser’s stain**, these granules appear **dark and intensely stained**, while the rest of the bacterial cell takes on a **green or blue** coloration.



Cultural Characteristics and Isolation Media:

A. Loeffler's Serum Medium:

An **enrichment medium** that contains horse or bovine serum.

Purpose:

It supports **rapid, luxuriant growth** of *Corynebacterium diphtheria* and enhances the development of its **distinctive club-shaped morphology** and **metachromatic (volutin) granules**.

This medium is especially useful for **early, preliminary diagnosis**, often providing recognizable growth within **6–8 hours of incubation**.

Loeffler's Serum for *Corynebacterium diphtheriae*



B. Tellurite Agar:

(e.g., Tinsdale's Agar or Hoyle's Agar)

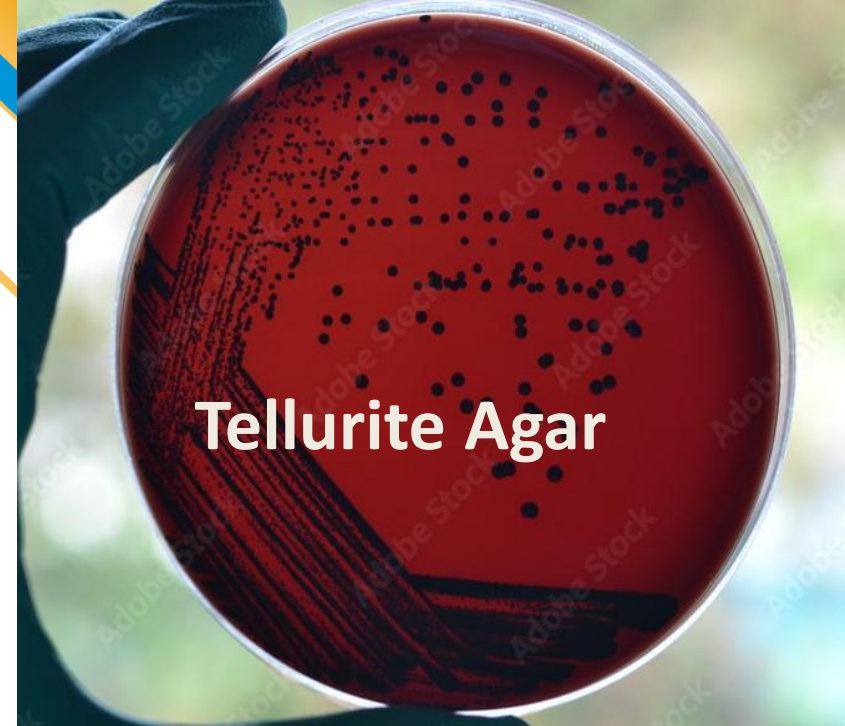
Key Component: Contains potassium tellurite.

Tinsdale Medium:

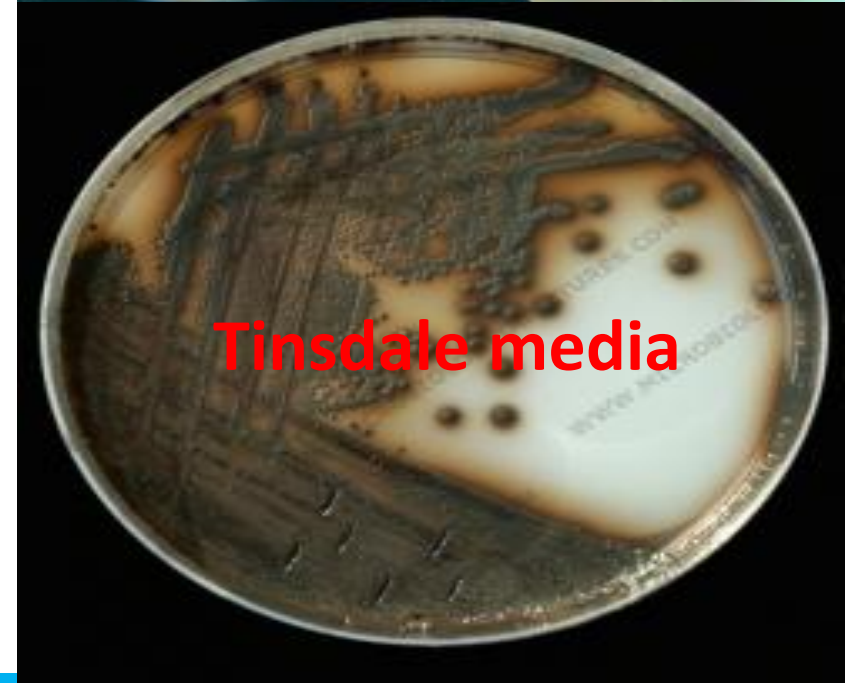
Produces grayish-black colonies surrounded by a dark brown halo, resulting from H_2S production a key diagnostic feature.

Colony Appearance:

Corynebacterium diphtheriae reduces tellurite to metallic tellurium, producing grayish-black to black colonies.



Tellurite Agar



Tinsdale media



C. diphtheriae Biotypes: These biotypes are sub-classifications of *Corynebacterium diphtheriae* based on the **morphological characteristics** of their colonies when grown on **tellurite-containing media** such as Tinsdale or Hoyle agar, include:

1. Mitis Biotype

- Forms **small, round, convex, black colonies** on tellurite-containing media.
- This biotype is commonly isolated and has a smooth colony appearance.

2. Intermedius Biotype

- Produces **small, flat, gray colonies**.
- The name “Intermedius” reflects that it shows features **intermediate** between Mitis and Gravis biotypes.

3. Gravis Biotype

- Develops **large, irregular, gray colonies**.
- Colonies have a rougher appearance compared to Mitis and Intermedius.

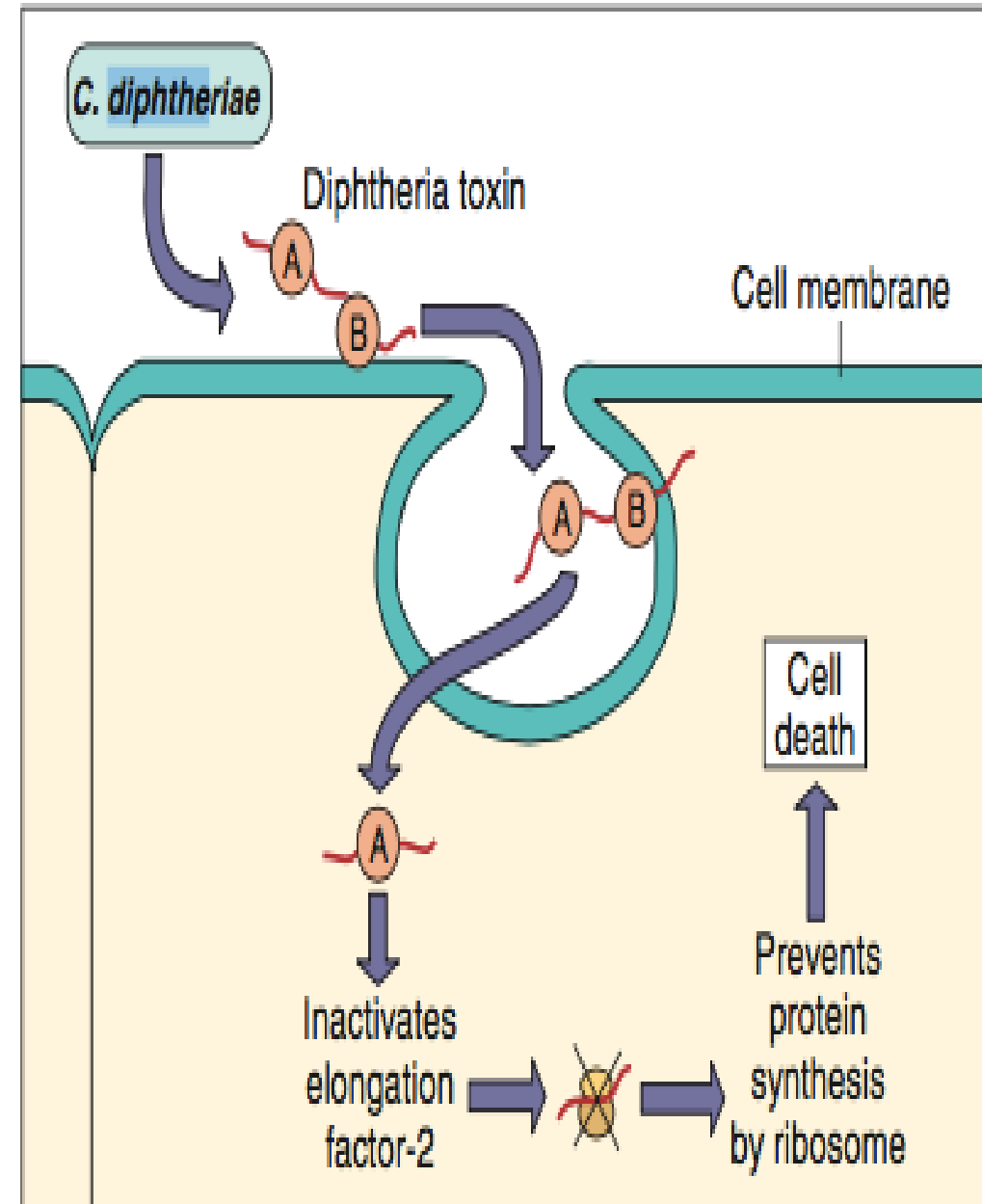
Virulence Factor(Diphtheria Toxin):

- A heat-labile Exotoxin.
- Inhibits protein synthesis in host cells.
- Expressed under low iron conditions.
- Consists of two fragments: A and B:

Fragment B: Responsible for binding and facilitating entry into the host cell.

Fragment A: The active component that blocks protein synthesis.

A Inhibition of protein synthesis



Diphtheria Toxin Detection (Toxigenicity Testing)

A. Elek Test (In Vitro):

Used to identify *C. diphtheriae* strains that produce diphtheria toxin, which is essential for severe disease.

Principle: Immunodiffusion test on agar.

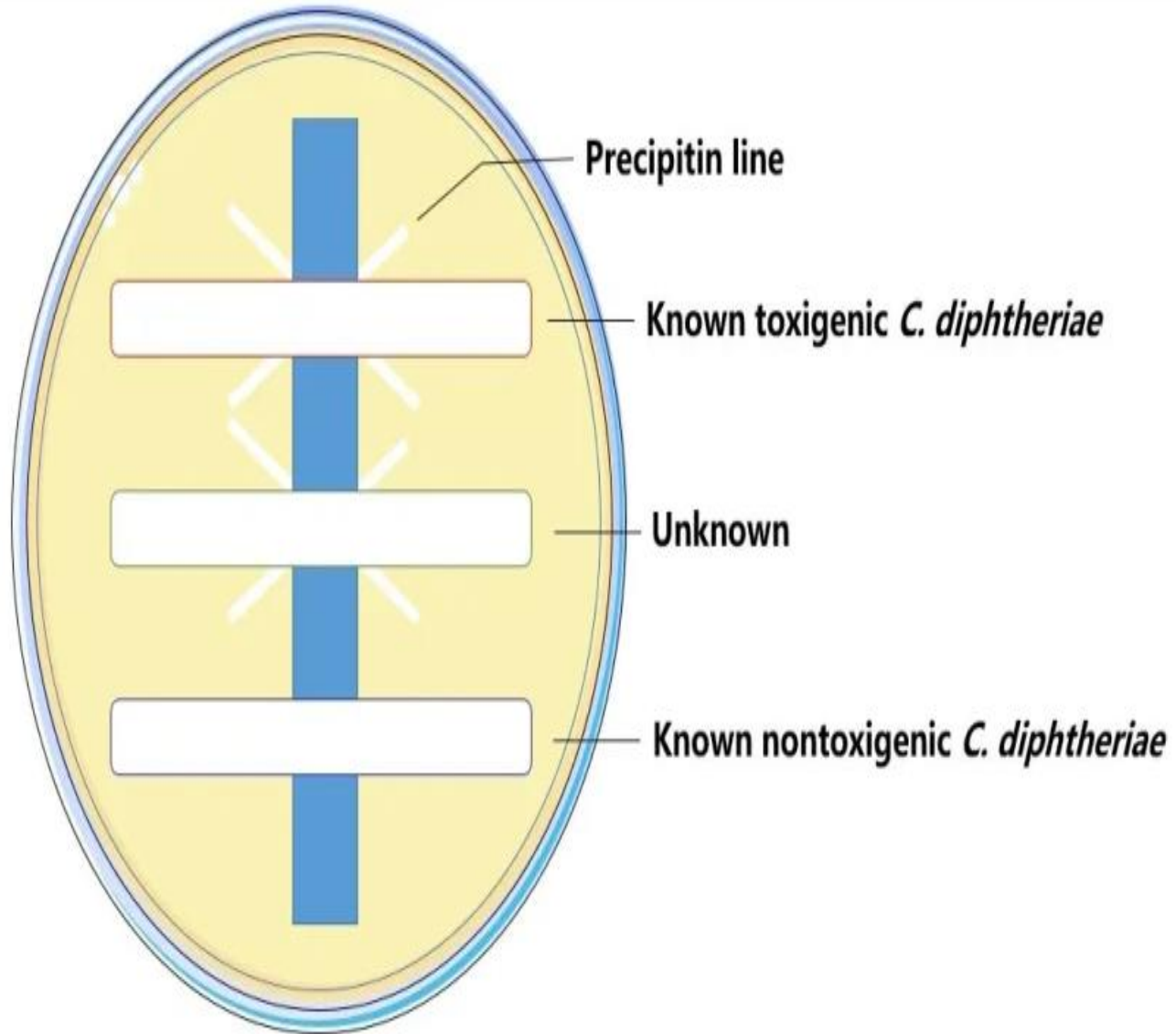
1-A filter strip soaked with diphtheria antitoxin is placed on the agar.

2-Suspected bacterial strains are streaked perpendicular to the strip.

3-Toxigenic strains release toxin (antigen) that diffuses and reacts with the antitoxin (antibody).

4-Formation of white precipitin lines at a 45° angle indicates toxin production.

ELEK's Test



Elek's test +ve
Toxogenic *C. Diphtheria*

B. Molecular Methods (PCR):

Principle: Detecting the presence of the Diphtheria toxin gene (tox gene) in the bacterial DNA using Polymerase Chain Reaction (PCR).

Advantage: Fast and highly sensitive.

C. Biochemical Tests

Catalase Test: Positive

Glucose Test: Positive (ferments glucose).

Urease Test: Negative



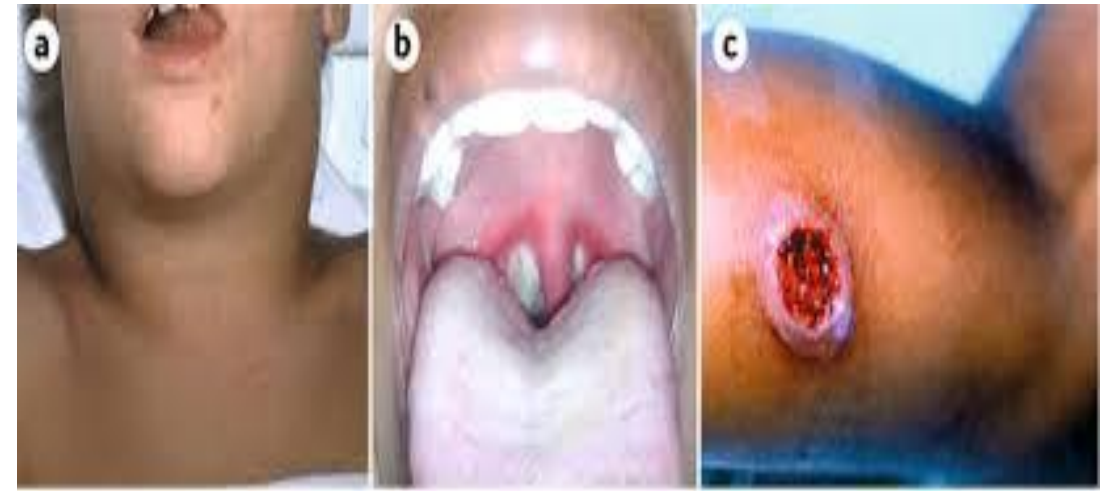
Pathogenesis:

Corynebacterium jeikeium

-**Opportunistic infections** (sepsis, endocarditis) in immunocompromised (e.g., patients with blood disorders, bone marrow transplants, intravenous catheters).

Corynebacterium urealyticum

-**Urinary tract infections (UTI's)**; cause chronic or recurrent cystitis, bladder stones, and pyelonephritis rare but important.





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Any Question?



Thank you